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Toda función holomorfa tiene primitiva holomorfa

Teorema 1 (Existencia de primitiva). *Sea* $f \in \mathcal{H}\left(D(z_0, r)\right)$

$$\implies \exists F \in \mathcal{H}\left(D(z_0, r)\right) : \forall z \in D(z_0, r) : F'(z) = f(z).$$

***Demostración*:** Para $z \in D(z_0, r)$ construimos un camino γ_z de z_0 a z tal que su imagen es la unión de los segmentos $\sigma_1 = [x_0 + iy_0, x + iy_0]$ y $\sigma_2 = [x + iy_0, x + iy]$. Tomamos $\sigma_1(t) = t + iy_0$ con $t \in [x_0, x]$ y $\sigma_2(t) = x + it$ con $t \in [y_0, y]$.

Consideramos también la "reflexión" $\widehat{\gamma_z}$ de z_0 a z tal que $\gamma_z \cup \widehat{\gamma_z} =: \partial R$. Para recorrer positivamente ∂R, en realidad tomamos $\partial R^+ = \gamma_z \cup (-\widehat{\gamma_z})$.

$$\overset{\substack{\text{C-G}\\ \downarrow}}{\implies} \int_{\partial R^+} f(w)\,\mathrm{d}w = \int_{\gamma_z} f(w)\,\mathrm{d}w - \int_{\widehat{\gamma_z}} f(w)\,\mathrm{d}w = 0 \implies \int_{\gamma_z} f(w)\,\mathrm{d}w = \int_{\widehat{\gamma_z}} f(w)\,\mathrm{d}w.$$

Definimos $F(z) := \int_{\gamma_z} f(w)\,\mathrm{d}w = \int_{x_0}^{x} f(t + iy_0)\,\mathrm{d}t + i\int_{y_0}^{y} f(x + it)\,\mathrm{d}t$

$$\implies F(z) = \int_{\widehat{\gamma_z}} f(w)\,\mathrm{d}w = i\int_{y_0}^{y} f(x_0 + it)\,\mathrm{d}t + \int_{x_0}^{x} f(t + iy)\,\mathrm{d}t.$$

$$\left.\begin{aligned} F_y(z) &= i\frac{\partial}{\partial y}\left(\int_{y_0}^{y} f(x+it)\,\mathrm{d}t\right) \overset{\substack{\text{RB}\\ \downarrow}}{=} if(x+iy) = if(z) \\ F_x(z) &= \frac{\partial}{\partial x}\left(\int_{x_0}^{x} f(t+iy)\,\mathrm{d}t\right) \overset{\substack{\text{RB}\\ \downarrow}}{=} f(x+iy) = f(z). \end{aligned}\right\} \implies F_x(z) = f(z) = -iF_y(z).$$

Luego F cumple las ecuaciones de Cauchy-Riemann. Como además f es continua en $D(z_0, r)$, se tiene que $F \in \mathcal{H}(D(z_0, r))$ y $F'(z) = F_x(z) = f(z)$. ∎

Referenciado en

- `teo-cauchy-goursat-disco`

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